Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

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eAppendix 1 Behavioral Counseling

All smokers received individual 15-minute behavioral smoking cessation counseling sessions conducted over 2 in-clinic and 3 telephone visits in Phase 1, and 2 in-clinic and one telephone visit in Phase 2. Relative to the initial target quit date (TQD; 7-days post-randomization), the 2 in-clinic counseling sessions in Phase 1 occurred 7 days prior and 34 days after the TQD, the latter being the point of re-randomization for Phase 2. The three phone sessions occurred at 1, 24-, and 30-days post TQD. Relative to the rerandomization visit, the one phone and 2 in-clinic visits for Phase 2 occurred 7, 21, and 42 days, respectively, post-rerandomization. Medication checks were conducted by phone (5 minutes) one day before and 3 after the TQD in Phase 1, and 3 and 14 days after re-randomization in Phase 2. The counseling protocol was manual driven (available from author) and has been used in several of our previous studies [1–3]. Counseling content involved preparation for quitting, identification of high risk situations for smoking, development of coping skills and direct support before and after the quit date, motivational intervention [4] for keeping or resetting a quit date, management of withdrawal symptoms, and medication compliance. All counselors were trained as certified tobacco treatment specialists.

eAppendix 2 Sample Size Estimation

Based on available data at the time this study was initiated[16], we did not expect differences in abstinence between the varenicline and NPL conditions at the end of Phase 1. However, we evaluated the differences between Phase 1 treatments to provide appropriate clinical information regarding the initial treatment choice. Anticipated probabilities of response at week 12 conditional on response/nonresponse at week 6 and subsequent continue or re-randomization were not directly available since a SMART trial like this one had never been done before. As such, we derived the anticipated probabilities for each of the cells at EOT from data from available meta-analyses[16] evaluating 12-week continuous abstinence outcomes, since 7-day point prevalence information (our primary outcome) was not available. Smoking cessation trials evaluating a new pharmacological treatment have often used continuous abstinence over the last 4 weeks of treatment as the primary outcome. However, seven-day point prevalence abstinence was selected as our primary outcome as it is most appropriate for this study design given that the second phase of treatment is 6 weeks in duration in contrast to typical pharmacological treatment trials evaluating 12 weeks of the same medication (e.g. dual nicotine replacement or varenicline). Nevertheless, using continuous abstinence for our initial estimates does provide a more conservative approach since continuous abstinence is typically lower than 7-day pp at similar end points.

We also took the additional step of modeling a range of potential outcomes as opposed to a single point estimate for abstinence at our primary endpoint. Given the uncertainty inherent in pre-existing empirical evidence and even greater uncertainty for conditional probabilities of response based on clinical consensus and marginal estimates, we elected to model probability distributions of the plausible effects (approximately +/- 2-17% points above or below the point estimates for abstinence rate within each condition) rather than simple point-estimates. This approach allows us to account for the uncertainty around any point estimate in our sample size calculations including the expected effects and the anticipated precision of these estimates. Type I Errors, a Frequentist statistical concept, are less applicable in Bayesian inference. Instead, the interpretation of the posterior probability is most salient: if a treatment demonstrates 95% posterior probability of benefit, this means that there is a 5% chance that it will result in a lack of benefit, or even in harm with the range of effect sizes covered by the posterior distribution indicating the degree to which this is the case.

E-Figure 1 provides our initial prior point estimates and 95% credible intervals for the anticipated probability of abstinence at 6 and 12 weeks, for each respective condition. These parameters correspond with the beta distributions specified for each outcome ascertainment measurement. Monte Carlo simulations of clinical trial outcomes incorporating sampling from these distributions was used to estimate the probability of detect effects, while accounting for the uncertainty in effect size estimates. We used logistic regression to analyze the data generated from the specified ~Beta distributions for “power” estimates, using vague, neutral priors for the coefficients in the log form (~N(μ = 0, σ = 1000).

eAppendix 3 Inverse Probability Weighting Data Modeling and Prior Specification

Inverse probability weighting permitted unbiased effect size estimates in the context of re-randomization. The inverse probability weights are based on the probabilities associated with the randomization steps. Given that these are established by design, they are known quantities and set as fixed. The probability of randomization the Phase I treatments is p = 0.5. The randomization to the second-phase treatments for Phase I abstainers is a probability of p = 1: they automatically remain on the same treatment. The non-abstinent participants at the end of Phase II are re-randomized to the continue, increase or switch conditions with probability p = 0.33. The inverse probability weights for the Phase I analysis (i.e. six weeks) are 1/0.5 = 2.0. For the Phase II analysis (i.e. twelve weeks) the inverse probability weights for the participants abstinent in Phase I are 1/(0.5*1) = 2. Analysis at the end of Phase II (i.e. twelve weeks) of participants who were non-abstinent at the end of Phase I and re-randomized to continue, increase or switch were 1/(0.5*0.33) = 6.06. These inverse probability weights permitted recovery of the anticipated parameter values as demonstrated in the simulations discussed in E-Table 1 and E-Table 2.

There are several discussions of inverse-probability weighting for SMART designs when only non-responders are rerandomized[5–7]. Implementation of the models used R v. 4.3 using the brms package[8–10]. Specifically, the weights are incorporated into the log-likelihood at the level of each observation: "Internally, this is implemented by multiplying the log-posterior values of each observation by their corresponding weights[11].

The coding procedures permitted identification of each salient effect. Specifically, at the initial randomization varenicline and NPL was coded as 1 and -1 respectively. For the re-randomization: 1) non-responders receiving continuation therapy were coded 1 while all other participants were coded 0; 2) non-responders receiving augmentation were coded 1 versus 0 for all other participants; 3) non-responders re-randomized to switching treatment were coded -1 versus 0 for all other participants.

<u>Data Analytic Models</u>

Our primary analytical approach using Bayesian inference permits the assessment of the proposed alternative hypotheses; estimates not available via a conventional Frequentist method. While Bayesian approaches permit direct estimates of the probability that the alternative hypothesis is true given the data, the Frequentist approach estimates the probability of the current data or data more extreme given that the null hypothesis is true[12]. This approach maps well onto clinical reasoning, has been suggested by the FDA as an approach for improving methodological efficiency conducting clinical research, and can provide probabilistic estimates that an effect exists even with smaller cell sizes[13–15].

This is particularly salient in the context of a SMART design where rerandomizations may result in small final cell sizes, and the focus must be on conducting the largest feasible trial. Finally, posterior distributions may form the basis of informative prior distributions for continued monitoring of treatments and treatment strategies exhibiting initial promise.

Logistic regression will utilize the following model for evaluating abstinence at twelve weeks:

$$logit(y)=\beta 0+\beta 1*a1+\beta 2*anr21+\beta 3*anr22+\beta 4*anr23$$

$$+\ \beta 5*a1*anr21+\beta 6*a1*anr22+\beta 7*a1*anr23$$

Where a1 is the indicator variable for initial treatment randomization, and anr21, anr22, anr23 represent the coded vectors for non-responder continuation, increase and switching respectively. Effects for each outcome cell at twelve weeks are as follows:

$\mathrm{C}NRT responders=\beta 0-\beta 1$

$Var\ responders=\beta 0+\beta 1$

$\mathrm{C}NRT nonresponders\rightarrow \mathrm{CN}RT=\beta 0-\beta 1+\beta 2-\beta 5$

$\mathrm{C}NRT nonresponders\rightarrow \mathrm{C}NRT+=\beta 0-\beta 1+\beta 3-\beta 6$

$\mathrm{C}NRT nonresponders\rightarrow Var=\beta 0-\beta 1-\beta 4+\beta 7$

$Var\ nonresponders\rightarrow Var=\beta 0+\beta 1+\beta 2+\beta 5$

$Var\ nonresponders\rightarrow Var+=\beta 0+\beta 1+\beta 3+\beta 6$

$Var\ nonresponders\rightarrow \mathrm{C}NRT=\beta 0+\beta 1-\beta 3+\beta 7$

Initially specified prior distributions for comparison of proportions use ~Beta(a = 1, b = 1) priors. Linear, logistic and Cox Proportional Hazards regression coefficient priors will take the form ~N (mean = 0, var = 1 x 10^6) in the linear, log (odds) and log scales respectively. Level one error variances will be specified as ~Inverse Gamma (shape = 0.001, scale = 0.001); level two variances will use ~Uniform(0,1000) distributions.

For all coefficients (intercept and treatment effects) we follow Spiegelhalter[16] and colleagues specifying vague/non-informative neutral priors using a ~Normal (mean = 0, variance = 1 x $10^{6)}$ on the log-scale. This encompasses an extremely broad range of priors: 95%CrI for the log-form = -1960 – 1960. Exponentiating these values would provide the 95% CrI for the odds ratio; extremely small/large limits. The advantage of using this prior is that while it is proper (integrates to 1), it is so broad as to be essentially flat across the range of potential values.

In the course of model-fitting we encountered difficulties in obtaining MCMC convergence. Inspection of the models suggested that the priors were so vague/non-informative that we obtained non-interpretable results for one of the interaction terms (e.g. odds ratio of 3.60 x 10^{239}). To address the non-convergence we revised the prior distribution for the intercept to ~Normal(mean = 0, variance = 10) in the log-form. For reference, if we estimate the prior 95% CrI for the odds represented by the intercept it is 3.07 x 10^{-9} – 325,215,956. Making this change facilitated convergence. We should note, that this interval is so broad as to have the same property as the original prior rendering it essentially flat in the region of any credible parameter value. Importantly, the priors for the treatment coefficients remained the same at ~Normal (mean = 0, variance = 1 x 10^{6}) on the log scale. Posterior predictive checking showed that the model continued to adequately fit the data. In essence we exchanged one vague/non-informative prior for another.

Based on the data available to us at the time, we hypothesized that relative to staying on the same medication, for varenicline non-abstainers, increasing the dose (varenicline+) would provide the best alternative, whereas for NPL non-abstainers, switching to varenicline would be the best subsequent treatment approach. For each hypothesis and comparisons to continuation, we estimated the probability of detecting any benefit (H_A: O.R. > 1) with a probability of 0.80, 0.85, 0.90 and 0.95, using K=1000 simulations (see **E-Table 1** and **E-Table 2**). All superiority hypotheses demonstrated a $\geq$ 80% chance of detecting 0.80 probability of benefit for a sample of N=500. **E-Table 2** also displays the point-estimates and 95% Credible Intervals of specified effects averaging over the K = 1,000 simulations. These point estimates and credible interval estimates derive from the proposed, data-analytic, logistic regression models. The average 95% Credible Intervals provide an index of the precision with which the current design may render estimates. In terms of clinical meaningfulness, meta-analytic review estimates the absolute risk reduction between two widely accepted first-line treatments (i.e., varenicline and bupropion) is 0.089 which is smaller than the lower-bound 95% Credible Limits for all contrasts in E-Table 2. As a pre-specified outcome, we stipulated a posterior probability of $\geq$ 0.80 that the postulated effect exists (i.e., Pr (O.R. > 1|data) $\geq$ 0.80) would constitute confirmation of our hypotheses, sufficient for informing clinical decision making and conducting future trials.

eAppendix 4 Supplemental Analyses for Secondary Outcomes of Continuous Abstinence at EOT+30 and 6 months

Secondary outcomes of continuous abstinence (CA) at EOT+30 and 6 months are described below. The study was not powered to detect differences using frequentist methods at these distal timepoints for continuous abstinence. As is the case in most smoking cessation research, a decline in absolute abstinence rates was observed over longer periods of follow-up after treatment had concluded.

Secondary Outcomes Summarized Analysis

The posterior probabilities and 95% Credible intervals for each treatment condition for CA at EOT+ 30 and 6-month follow-up points are provided in **E-Figures 2 & 3**, respectively**.** The same contrasts as carried out for the 7-day pp outcome at EOT were conducted at both time points, calculating the probability of a non-zero difference for each contrast, as a metric for the benefit for switching and increasing approaches, relative to continuing, for week 6 non-abstainers initially treated with either CNRT or varenicline. The absolute risk difference (ARD) and 95% CrI between the comparison of each continue condition to the respective switch and increase dose conditions within each of the phase 1 non-abstainer groups treated with either CNRT or varenicline are presented in **E-Table 9.** In addition, the contrast between the respective switch and increase conditions within CNRT and varenicline [phase 1 non-abstainer groups are presented in **E-Table 10**. Similarly, the contrast between phase 1 abstainers (those abstaining at week 6 and remaining of the same medication) in the CNRT and varenicline conditions at each of the follow-up time points are presented in **E-Table 11**.

Posterior probability distributions for CA at EOT+30 are shown in **E-Figure 2**. Among CNRT Phase 1 non-abstainers (n=191), both the dose increase to CNRT+ (8%; n=50) or switching to varenicline (10%; n=51) provided more benefit (posterior probability >99%; see **E-Table 9** for ARD and 95% Crl) than continuing CNRT (3%; n=90). However, for

Phase 1 non-abstainers on varenicline (n=157), only varenicline+ (8%; n=39) provided benefit relative to continuation (0%; n=42). For 6-month CA (**E-Figure 3**) among both CNRT and varenicline Phase 1 non-abstainers, only the respective dose increase condition (CNRT+ 3%; varenicline+ 2%) provided benefit over continuation (0%) with probabilities of 96% and 99%, respectively of a non-zero difference (see **E-Table 9). E-Table 10** also provides the ARD and probabilities for the phase 2 increase vs. switch contrasts for both CNRT and varenicline phase 1 non-abstainers at each timepoint. Among Phase 1 abstainers, at EOT+30 a higher proportion of CA was observed for those on CNRT than those on varenicline (probability 97%), but no differences were observed at 6 months (see **E-Figures 2** and **3** and **E-Table 10**).

Secondary Outcome Detailed Analysis

<u>Continuous Abstinence at EOT+30 for CNRT Non-Abstainers</u>

Among participants initially randomized to CNRT who were non-abstinent at six weeks, switching to varenicline, increasing the CNRT dose (CNRT+: 2 x 21 mg nicotine patches plus lozenge), or continuation on the same dose of CNRT (21 mg patch plus lozenge) resulted in probabilities and 95% Crl of abstinence at EOT+30 of 1.0% (7.0%-1.3%), 8.0% (5.0%-1.1%) and 3.0% (2.0%-5.0%) respectively (see **E-Figure 2**), with a benefit for increasing the dose (CNRT+) or switching to varenicline as compared to continuation. Contrasting participants remaining in CNRT continuation with CNRT+ and switching to varenicline demonstrated a posterior probability of >99% for the benefit of increasing the dose (CNRT+): ARD (95% Crl) = 5.0%, (1.0%-8.0%) and a posterior probability of >99% for the benefit of switching to varenicline: ARD = 6.0% (3.0%-1.0%) (see **E-Table 9**). Further, contrasting CNRT+ and switching to varenicline provided a posterior probability of 79% for the benefit of switching: ARD = -2.0% (-6.0%-3.0%) (see **E-Table 10)**. This suggests that among those initially assigned to CNRT and non-abstinent at week 6, the chance of continuous abstinence at EOT+30 is higher for the switching strategy (to varenicline) than for the increasing the dose to CNRT+ with CNRT continuation being the lowest.

<u>Continuous Abstinence at EOT+30 for Varenicline Non-Abstainers</u>

Among participants initially randomized to varenicline who were non-abstinent at 6 weeks, increasing the dose (varenicline+), switching to CNRT, or continuation on varenicline resulted in probabilities of continuous abstinence at EOT+30 of 8.0% (5.0%-1.1%), 0.0% (0.0%-0.0%) and 0.0% (0.0%-0.0%) (see **E-Figure 2**) respectively, with a benefit for varenicline+ as compared to continuation or switching. Contrasting varenicline+ with varenicline continuation demonstrated a posterior probability of >99% for the benefit of varenicline+: ARD =8.0% (5.0%-1.1%) (see **E-Table 9**). Similarly contrasting varenicline+ with switching to CNRT showed an ARD = 8.0% (5.0%-1.1%) and posterior probability of benefit of > 99%. This suggests that among smokers initially assigned to varenicline, who were non-abstinent at week 6, increasing varenicline to 3 mg for another 6 weeks (to week 12) is better than switching to CNRT or continuation for improving the chance of continuous abstinence 30 days after EOT, with an estimated increase in absolute abstinence rates of approximately 8%.

<u>Continuous Abstinence at EOT+30 for Phase 1 Abstainers Continuing Medication</u>

Among participants initially randomized to CNRT and varenicline who were abstinent at 6 weeks (not re-randomized), continuation on these treatments resulted in probabilities of continuous abstinence at EOT+30 of 67% (58%-75%) and 56% (48%-63%), respectively (see **E-Figure 2**). The posterior probability of a difference between these two conditions was 97%: ARD = 1.1% (-1.0%-22%) for the benefit of CNRT continuation among CNRT abstainers (see **E-Table 11**).

<u>Continuous Abstinence at 6 months for CNRT Non-Abstainers</u>

Among participants initially randomized to CNRT who were non-abstinent at six weeks, switching to varenicline, or increasing the CNRT dose (CNRT+) (2 x 21 mg nicotine patches plus lozenge), or continuation on the same dose of CNRT (21 mg patch plus lozenge) resulted in probabilities of continuous abstinence at 6 months of 4% (2.0%-6.0%), 6.0% (4%-9%) and 3.0% (2.0%-5.0%) respectively (see **E-Figure 3**). Contrasting participants remaining in CNRT continuation with CNRT+ and switching to varenicline conditions demonstrated a posterior probability of 96% for the benefit of CNRT+: ARD = 3.0% (0.0%-6.0%) and a posterior probability of 66% for the small benefit of switching: ARD = 1.0% (-2.0%-3.0%) (see **E-Table 7**). Further, contrasting CNRT+ and switching to varenicline provided a posterior probability of 88% for the benefit of CNRT+: ARD = 2.0% (-1.0%-6.0%) (see **E-Table 10**). This suggests that among those initially assigned to CNRT and non-abstinent at week 6, the chance of CA at 6 months is slightly higher for CNRT+ vs continuation.

<u>Continuous Abstinence at 6 months for Varenicline Non-Abstainers</u>

Among participants initially randomized to varenicline who were non-abstinent at 6 weeks, increasing the dose (varenicline+), switching to CNRT, or continuation on varenicline resulted in probabilities of abstinence at 6 months of 2.0% (1.0%-5.0%), 0.0% (0.0%-0.0%) and 0.0% (0.0%-0.0%) respectively (see **E-Figure 3**), with a benefit for varenicline+ compared to continuation or switching. Contrasting varenicline+ with varenicline continuation and switching to CNRT demonstrated a posterior probability of >99% for the benefit of varenicline+: ARD = 2.0% (1.0%-5.0%) for the benefit of varenicline+ relative to both continuation on varenicline (see **E-Table 9**) and switching to CNRT (see E**-Table 9**), with no benefit: ARD=0.0% (0.0%,0%) for CNRT relative to continuation. This suggests that among smokers initially assigned to varenicline, who were non-abstinent at week 6, increasing varenicline to 3 mg for another 6 weeks (to week 12) is better than switching to CNRT or continuation for the chance of continuous abstinence at 6 months, with an estimated increase in absolute abstinence rates of approximately 2%.

Continuous Abstinence at 6 months for Abstainers

Among participants initially randomized to CNRT and varenicline who were abstinent at 6 weeks, continuation on these treatments resulted in probabilities of abstinence at 6 months of 39% (30%-48%) and 40% (33%-47%) (see E-Figure 3) respectively. The posterior probability of a difference between these two conditions was 55%, ARD = 1.0% (-1.3%-1.1%) indicating essentially no difference for the small benefit of varenicline continuation.

eAppendix 5 Post-hoc Analyses for Dynamic Treatment Effects

At the request of the statistical reviewer we have estimated the dynamic treatment effects (DTE's) for our primary outcome following approaches used in the literature that rely on inverse-probability weighting to account for re-randomization of non-responders only, and data augmentation to permit estimates that incorporate responders into each dynamic treatment regimen[5–7]. Estimation of these DTE's were not a part of our original statistical plan, but we do so here to provide interested readers with information regarding average treatment effects across an entire pathway (Phase1 and phase 2 results combined). Of note, the recommended approach for this applies a generalized estimating equation (GEE) with robust standard errors. We took this frequentist approach because at present there does not yet seem to be a clear Bayesian analogue for use in our current design which requires incorporation of inverse probability weighting [8,9]. Similarly, the approach developed for small-n SMART designs does not currently have the flexibility necessary for our setting [10,11].

E-Table 12 reports the point estimates and 95% confidence intervals derived from the Frequentist GEE for the average probability of abstinence at week 12 of each treatment pathway combining the phase 1 treatment responders with non-responders receiving one of the three alternatives in phase 2 (stay, switch, increase dose).

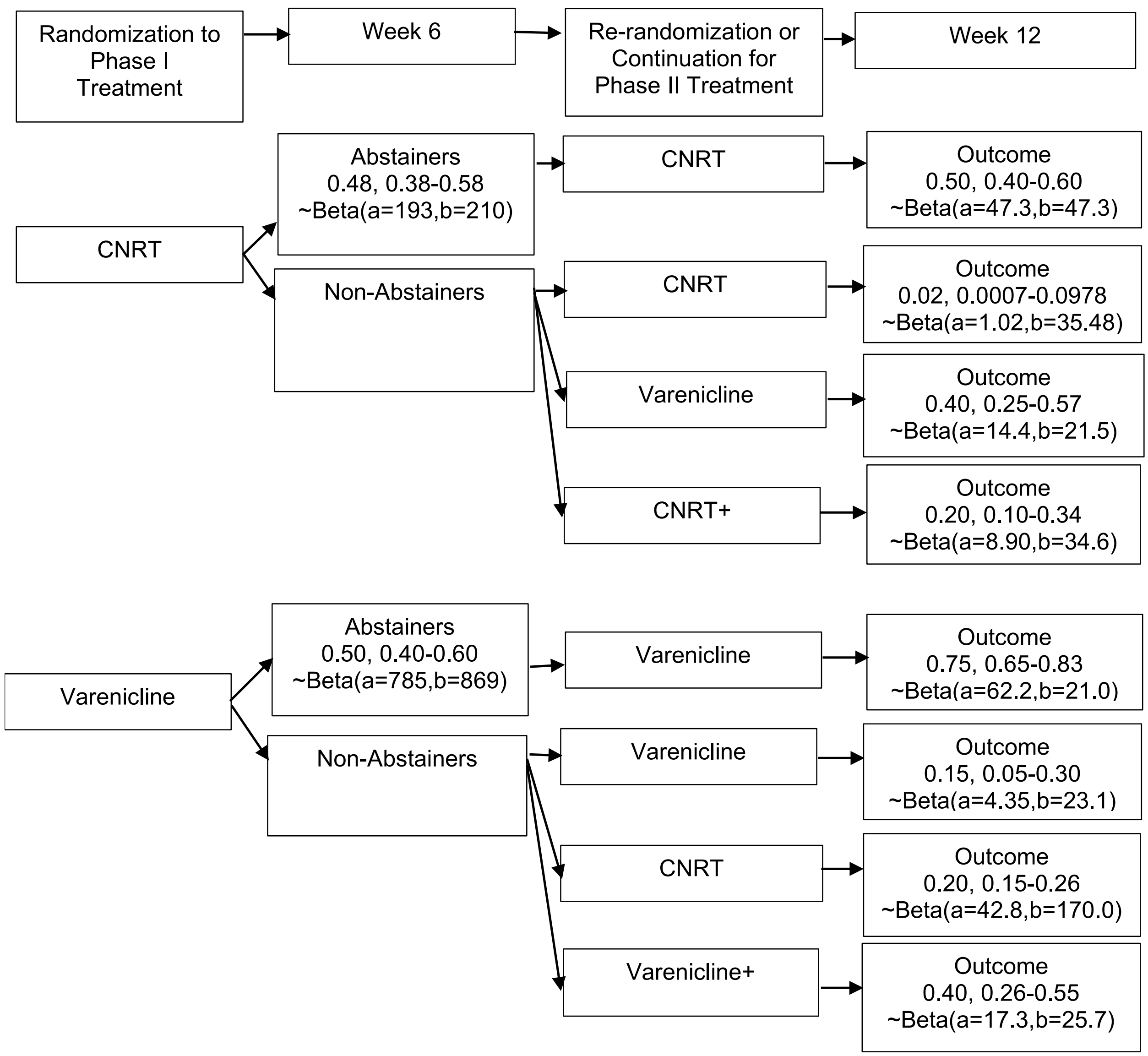
It is important to note that unlike Bayesian credible intervals the confidence intervals do not possess a distribution across it the way that the Bayesian posterior does, and does not allow discussion of the probabilities of an alternative hypothesis[12].

Taking the perspective of clinical decision making in today's clinical environment where both CNRT and varenicline remain viable first-line treatments, for which the choice may be based on cost, availability, previous experience, patient/provider preference, in addition to efficacy, comparisons within initial treatment arms best fit the current clinical milieu. As shown in **E-Table-12,** the results of combining Phase 1 treatment abstainers remaining on an initial treatment with non-abstainers rerandomized to one of the three alternatives are consistent with those reported for our primary outcome comparing each stay condition in phase 2 with the corresponding switch and dose increase condition within that primary treatment arm. The highest abstinence for those starting on varenicline was achieved when combining week-6 abstainers with non-abstainers receiving a dose increase (varenicline +). For those starting on CNRT, similar abstinence rates were observed for the combination including week-6 abstainers remaining on CNRT with non-abstainers in the CNRT+ or switch (varenicline) conditions. Hence, this post-hoc analyses as well as our planned contrasts suggest that increasing the dose of varenicline for those starting on varenicline but unable to abstain, would be the best alternative treatment approach, as will either increasing the dose of CNRT, or switching CNRT non-abstainers to varenicline.

Tables and Figures

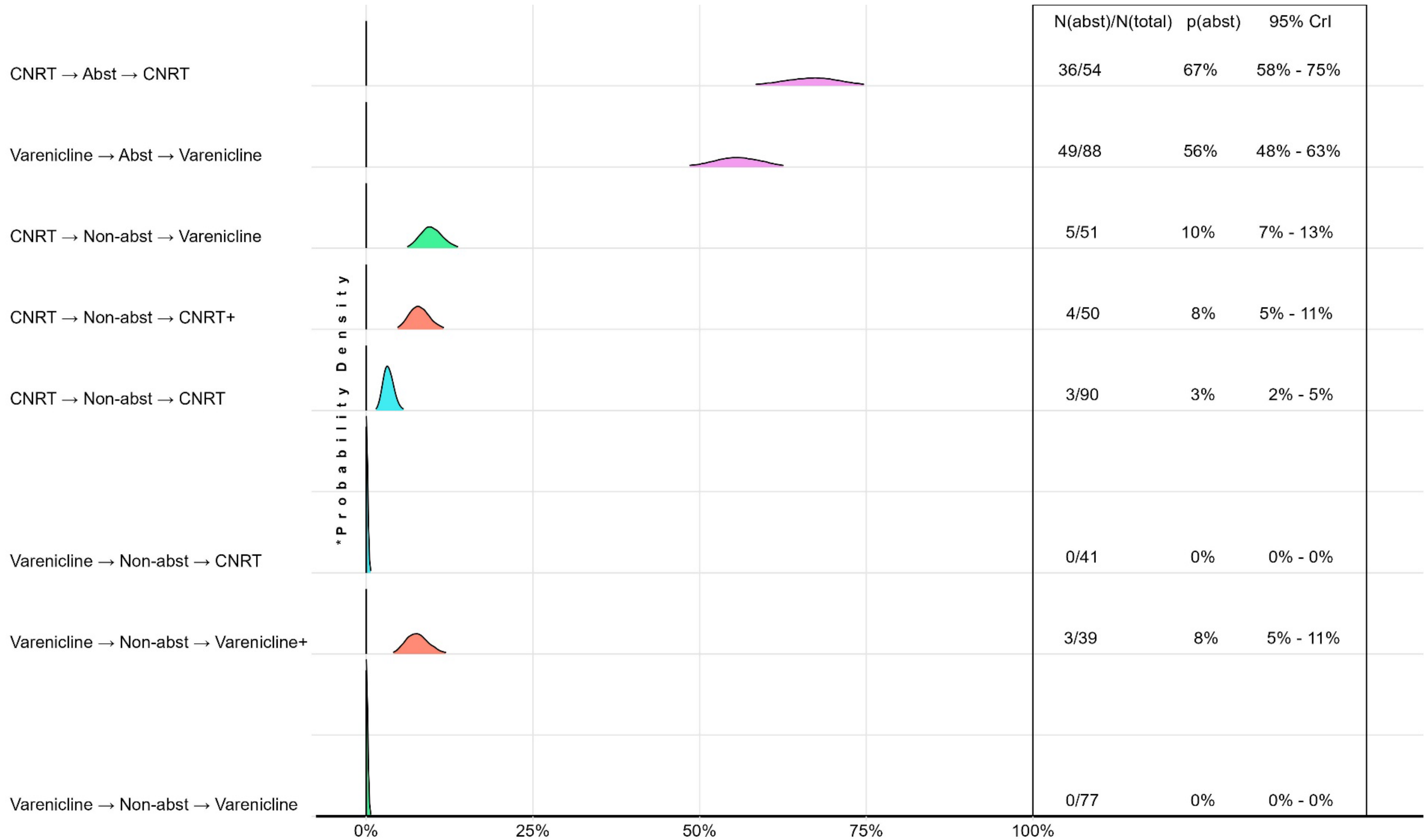
E-Figure 1. Anticipated Probabilities for 7-day point prevalence at EOT (Primary Outcome).

Smokers were initially randomized to receive either active Varenicline or active Nicotine Patch plus lozenge (**CNRT**) and assessed for abstinence at week 6. Abstainers continued the same medication. Non-abstainers were re-randomized to either continue the same medication, switch to the alternate medication, or increase the dose of the current one (varenicline 2 mg to 3 mg [**Varenicline**+] and increasing patch dose to two 21mg patches + 4 mg lozenge [**CNRT**+]).



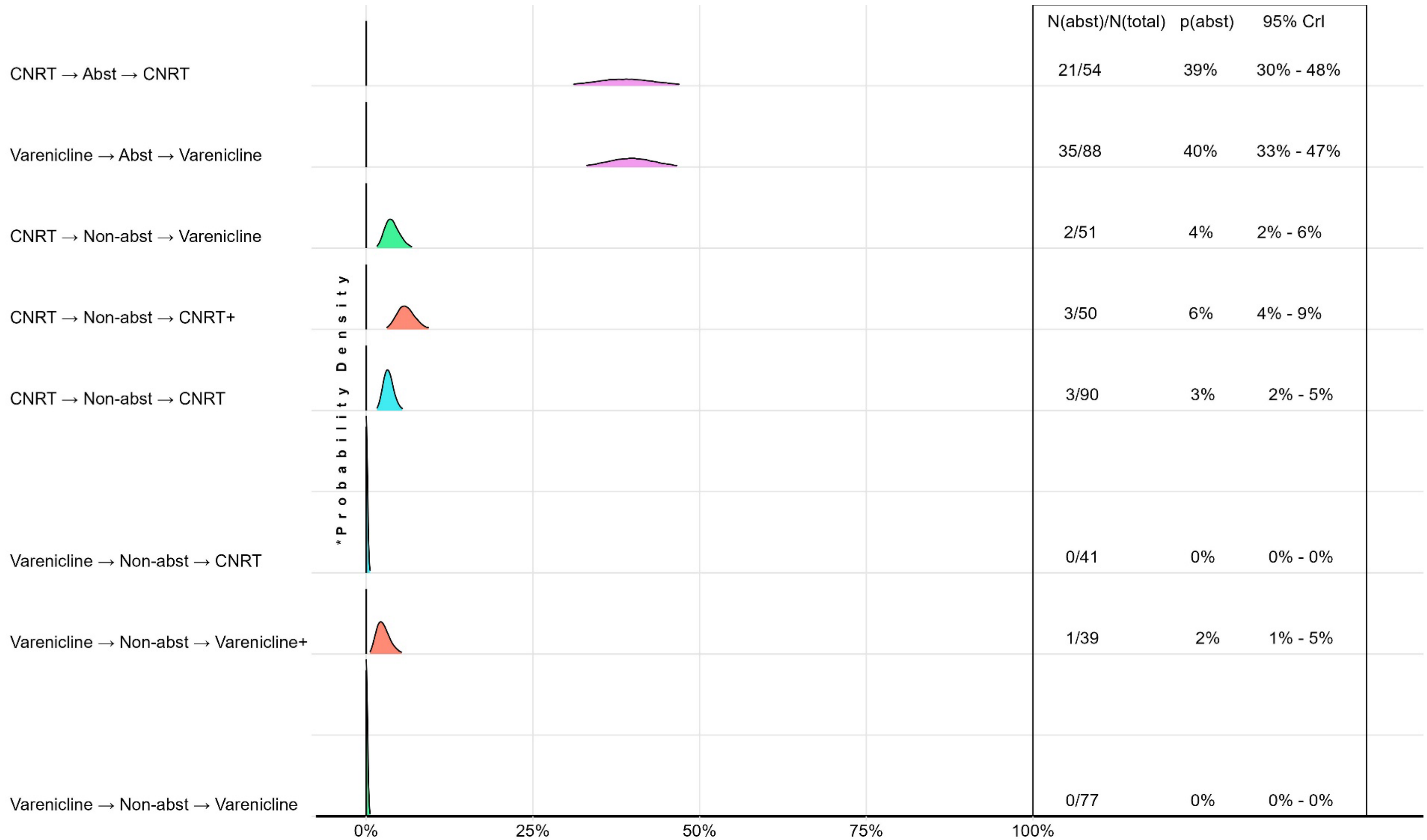
E-Figure 2. Posterior probability distributions, point estimates and 95% credible intervals for the probability of Continuous abstinence at the EOT (week 12) + 30 days follow-up as a function of phase 1 treatment condition, and end of phase 1 smoking status.

*Each distribution is a probability density with an area under the curve equal to one. The height (y) of the density is scaled to permit full depiction of each distribution. Labels for each curve include: Phase 1 drug (CNRT or Varenicline) at the leftmost position, followed by an indicator (→) of abstinence status at week 6 for participants in that cell (Abstinent=Abst; Non-abstinent= Non-abst.), followed by an indicator (→) of the drug assignment for phase 2 (CNRT, Varenicline, Varenicline+ [dose increase to 3 mg/day] or CNRT+ [dose increase to two 21mg patches plus 2mg lozenge]). The columns to the right of each curve provide cell sample size, estimated probability of abstinence (p(Abst)) and the corresponding 95% Credible Interval (95% Crl).



E-Figure 3. Posterior probability distributions, point estimates and 95% credible intervals for the probability of Continuous abstinence at the 6-month post TQD (target quit-date) follow-up as a function of phase 1 treatment condition, and end of phase 1 smoking status.

*Each distribution is a probability density with an area under the curve equal to one. The height (y) of the density is scaled to permit full depiction of each distribution. Labels for each curve include: Phase 1 drug (CNRT or Varenicline) at the leftmost position, followed by an indicator (→) of abstinence status at week 6 for participants in that cell (Abstinent=Abst; Non-abstinent= Non-abst.), followed by an indicator (→) of the drug assignment for phase 2 (CNRT, Varenicline, Varenicline+ [dose increase to 3 mg/day] or CNRT+ [dose increase to two 21mg patches plus 2mg lozenge]). The columns to the right of each curve provide cell sample size, estimated probability of abstinence (p(Abst)) and the corresponding 95% Credible Interval (95% Crl).



E-Table 1. The probability point-estimates and 95% Credible Limits for each study cell, derived from logistic regression models, averaged across K = 1,000 simulations.

The proposed data analysis model extracted the specified parameters. CNRT=Combined Nicotine Replacement (21 mg patch + 2 mg lozenge); CNRT+ =(Two 21 mg patch + 2 mg lozenge); VAR= Varenicline 2mg/day; VAR+ = Varenicline 3mg/day

Effect	Average Point Estimate	Average Interval (95% C.I.) Estimate
Phase 1 Abstainers		
CNRT	0.504	0.441-0.566
VAR	0.749	0.692-0.801
Phase 2 Non-Abstainers		
CNRT→CNRT (continue)	0.029	0.017-0.047
CNRT→CNRT+ (increase)	0.204	0.160-0.255
CNRT→VAR (switch)	0.399	0.343-0.458
VAR→VAR (continue)	0.159	0.120-0.203
VAR→ VAR+ (increase)	0.404	0.347-0.463
VAR→ CNRT (switch)	0.201	0.157-0.251

E-Table 2. Estimates of probability of detecting any benefit (H_A: θ > 0) conferred by the treatment hypothesized as being superior. Power for detecting benefit at posterior probability levels at .80 or greater are provided for each comparison.

All superiority hypotheses demonstrate a $\geq$ 80% chance of detecting 0.90 probability of benefit for a sample of N = 500. Also shown are point-estimates and 95% Credible Intervals of specified effects averaging over the K = 1,000 simulations. CNRT=Combined Nicotine Replacement (21 mg patch + 2 mg lozenge); CNRT+ =(Two 21 mg patch + 2 mg lozenge); VAR= Varenicline 2mg/day; VAR+ = Varenicline 3mg/day

Effect	Posterior Probability Pr(θ>0)	Power to Detect Pr(θ>0)	Average Point Estimate	Average Interval (95% C.I.) Estimate
Effect of Treatment Phase I Abstainers	0.80	0.948	0.240	0.160-0.327
VAR>CNRT at EOT	0.85	0.980		
	0.90	0.974		
	0.95	0.963		
CNRT Phase I Non-Abstainers to rerandomized to Switch, Continue or Increase Conditions in Phase 2				
VAR (switch) > CNRT (continue)	0.80	0.999	0.370	0.309-0.431
	0.85	0.999		
	0.90	0.999		
	0.95	0.998		
CNRT+ (increase) > CNRT (continue)	0.80	0.964	0.175	0.125-0.228
	0.85	0.957		
	0.90	0.951		
	0.95	0.939		
VAR (switch) vs. CNRT +(increase)	0.80	0.878	0.195	0.119-0.269
	0.85	0.863		
	0.90	0.842		
	0.95	0.819		
VAR Phase I Non-Abstainers to rerandomized to Switch, Continue or Increase Conditions in Phase 2				
VAR+ (increase) vs. VAR (continue)	0.80	0.930	0.245	0.172-0.316
	0.85	0.922		
	0.90	0.915		
	0.95	0.895		
VAR+ (increase) vs. CNRT (switch)	0.80	0.917	0.202	0.127-0.288
	0.85	0.905		
	0.90	0.890		
	0.95	0.859		

E-Table 3. Baseline Measures and Demographics by Phase 1 Treatment by Phase 1 Response by Phase 2 Treatment.

Phase 1 Treatment	CNRT				Varenicline			
Phase 1 Response	Abstainer	Non-abstainer			Abstainer	Non-abstainer		
Phase 2 Treatment	CNRT (N=54)	VAR (N=51)	CNRT (N=90)	CNRT+ (N=50)	VAR (N=88)	CNRT (N=41)	VAR (N=77)	VAR+ (N=39)
Demographics								
Age	50.7 (11.0)	49.6 (10.8)	47.2 (11.7)	49.4 (11.7)	48.7 (11.3)	46.2 (11.4)	47.1 (11.2)	46.0 (10.9)
Sex, n (%)								
Female	21 (38.9)	21 (41.2)	41 (45.6)	22 (44.0)	38 (43.2)	22 (53.7)	33 (42.9)	12 (30.8)
Male	33 (61.1)	30 (58.8)	49 (54.4)	28 (56.0)	50 (56.8)	19 (46.3)	44 (57.1)	27 (69.2)
Race and ethnicity [a], n (%)								
African American, non-Hispanic	14 (25.9)	15 (29.4)	24 (26.7)	11 (22.0)	19 (21.6)	12 (29.3)	17 (22.1)	17 (43.6)
Asian	1 (1.9)	2 (3.9)	4 (4.4)	2 (4.0)	2 (2.3)	1 (2.4)	1 (1.3)	-
Hispanic, any race	5 (9.3)	5 (9.8)	6 (6.7)	3 (6.0)	8 (9.1)	3 (7.3)	10 (13.0)	1 (2.6)
More than one race	-	2 (3.9)	2 (2.2)	1 (2.0)	3 (3.4)	1 (2.4)	2 (2.6)	-
White, non-Hispanic	34 (63.0)	25 (49.0)	52 (57.8)	32 (64.0)	56 (63.6)	24 (58.5)	45 (58.4)	20 (51.3)
Other	-	2 (3.9)	2 (2.2)	1 (2.0)	-	4.9 (2)	2 (2.6)	1 (2.6)
Employment, n (%)								
Employed	72.2 (39)	74.5 (38)	70.0 (63)	80 (40)	75.0 (66)	73.2 (30)	71.4 (55)	69.2 (27)
Unemployed	27.8 (15)	25.5 (13)	30.0 (27)	18 (9)	25.0 (22)	26.8 (11)	28.6 (22)	28.2 (11)
Education, median (IQR)	13 (12 – 16)	13 (13 – 14)	13 (12 – 14)	13 (12 – 14)	14 (12.75 – 16)	13 (12 – 14)	13 (12 – 14)	13 (12 – 13)
Household Income, n (%)								
< $50,000	24 (44.4)	32 (62.7)	41 (45.6)	18 (36.0)	34 (38.6)	21 (51.2)	30 (39.0)	22 (56.4)
$50,000 - $79,999	8 (14.8)	11 (21.6)	21 (23.3)	11 (22.0)	20 (22.7)	9 (22.0)	29 (37.7)	10 (25.6)
> $80,000	22 (40.7)	8 (15.7)	25 (27.8)	18 (36.0)	31 (35.2)	10 (24.4)	15 (19.5)	7 (17.9)
Not reported	-	-	3 (3.3)	3 (6.0)	3 (3.4)	1 (2.4)	3 (3.9)	-
Smoking Characteristics								
CO [b], median (IQR), ppm	17 (10.25 – 23.5)	17 (12.5 – 28.5)	20 (13.25 – 26)	22.5 (15 – 32)	19.5 (12 – 27)	22 (19 – 36)	22 (16 – 28)	20 (13.5 – 25.5)

Phase 1 Treatment	CNRT				Varenicline			
Phase 1 Response	Abstainer	Non-abstainer			Abstainer	Non-abstainer		
Phase 2 Treatment	CNRT (N=54)	VAR (N=51)	CNRT (N=90)	CNRT+ (N=50)	VAR (N=88)	CNRT (N=41)	VAR (N=77)	VAR+ (N=39)
Cigarettes per day, median (IQR)	15 (11 – 20)	16 (10 – 20)	20 (12 – 20)	20 (15 – 25)	18.5 (10 – 20)	20 (11.75 – 25)	20 (15– 25)	15 (10 – 20)
FTCD [c], median (IQR), total score	4 (2 – 5)	5 (3 – 6)	6 (4 – 7)	5 (4 – 7)	4 (2 – 6)	6 (4 – 6.25)	5 (4 – 7)	5 (3 – 6)
Years of smoking, mean (SD)	29.6 (13.1)	28.6 (11.8)	27.7 (12.0)	29.4 (13.6)	27.6 (12.4)	28.0 (11.8)	28.2 (11.2)	28.2 (12.0)
Age of smoking initiation, median (IQR)	17.0 (15.0 – 20.8)	15.0 (13.0 – 19.5)	16.0 (15.0 – 18.8)	17.5 (15.0 – 19.0)	17.0 (14.0 – 20.0)	16.0 (14.0 – 19.0)	17.0 (15.0 – 18.0)	17.0 (15.0 – 18.0)

Abbreviations: CNRT=Combined Nicotine Replacement Therapy (21 mg nicotine patch plus 2 mg lozenge); CNRT+ = Two 21 mg patch + 2 mg lozenge; VAR= Varenicline 2mg/day; VAR+ = Varenicline 3mg/day; CO=Carbon Monoxide; ppm=parts per million; SD=Standard Deviation; IQR=Interquartile Range; FTCD=Fagerström Test for Cigarette Dependence [39].

[a] We used NIH categories for the self-reported race and ethnicity.

[b] A CO level of 6 ppm or higher was required for eligibility.

[c] A 6-item instrument with a possible range of 0 to 10 with higher scores meaning more cigarette dependence. In general, a score of 5 or higher is considered moderate to severe dependence.

E-Table 4 ARD and NNT* for Seven-Day Point Prevalence at EOT

Contrast	ARD	ARD 95%LCrI	ARD 95%UCrI	NNT
CNRT-Abst. vs. Varenicline-Abst.	6%	-4%	16%	16
CNRT-Non-Abst. -->CNRT+(increase) vs. -->CNRT (stay)	6%	2%	11%	16
CNRT-Non-Abst. -->Varenicline (switch) vs. -->CNRT (stay)	6%	2%	10%	17
CNRT-Non-Abst. -->CNRT+ (increase) vs. -->Varenicline (switch)	0%	-5%	6%	378
Varenicline-Non-Abst. -->Varenicline+ (increase) vs.-->Varenicline (stay)	18%	13%	23%	6
Varenicline-Non-Abst. -->CNRT (switch) vs. Varenicline-(stay)	3%	1%	4%	39
Varenicline-Non-Abst -->Varenicline+ (increase) vs. --> CNRT (switch)	20%	16%	26%	5
*NNT=Number needed to Treat				

E-Table 5. Number of participants (n) experiencing adverse events in Phase 1 by organ system and treatment. Point Estimates and 95% Credible Intervals are shown. In Phase 1, we examined the difference between VAR and CNRT. N=number of events; Estimate= estimated rates for N events, based on inverse-probability weighting described in our design.

There were no adverse events for which these differences exceeded 2%, except for nausea in VAR. Estimates should be considered descriptive with no correction for multiple comparisons.

System	**AE**	**CNRT**			**Varenicline**		
	N	**N**	**Estimate**	**95% CrI**	**N**	**Estimate**	**95% CrI**
CARDIOVASCULAR	Palpitations	3	1.49	0.44-3.52	0	0.28	0.01-1.49
DERMATOLOGIC	Dry Skin	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Hyperhidrosis	1	0.68	0.1-2.24	1	0.68	0.1-2.24
	Pain Of Skin	1	0.68	0.1-2.24	1	0.68	0.1-2.24
	Pruritus	20	8.39	5.36-12.28	13	5.55	3.15-8.87
	Skin Rash	15	6.36	3.76-9.86	6	2.71	1.15-5.23
EAR, NOSE & THROAT	Allergic Rhinitis	2	1.09	0.25-2.91	3	1.49	0.44-3.52
	Tinnitus	0	0.28	0.01-1.49	1	0.68	0.1-2.24
GASTROINTESTINAL	Abdominal Pain	0	0.28	0.01-1.49	2	1.09	0.25-2.91
	Bloating	2	1.09	0.25-2.91	2	1.09	0.25-2.91
	Constipation	5	2.3	0.9-4.68	10	4.33	2.25-7.35
	Diarrhea	10	4.33	2.25-7.35	14	5.95	3.45-9.36
	Dry Mouth	1	0.68	0.1-2.24	2	1.09	0.25-2.91
	Dyspepsia	9	3.93	1.97-6.83	6	2.71	1.15-5.23
	Flatulence	2	1.09	0.25-2.91	2	1.09	0.25-2.91
	Gastroesophageal Reflux Disease	0	0.28	0.01-1.49	2	1.09	0.25-2.91
	Hepatobiliary Disorders	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Hiccups	0	0.28	0.01-1.49	2	1.09	0.25-2.91
	Lip Pain	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Mucositis Oral	1	0.68	0.1-2.24	1	0.68	0.1-2.24
	Nausea	17	7.17	4.39-10.83	54	22.19	17.31-27.65
	Oral Dysesthesia	3	1.49	0.44-3.52	1	0.68	0.1-2.24
	Periodontal Disease	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Stomach Pain	2	1.09	0.25-2.91	3	1.49	0.44-3.52
	Thirst	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Toothache	2	1.09	0.25-2.91	0	0.28	0.01-1.49
	Vomiting	1	0.68	0.1-2.24	1	0.68	0.1-2.24
GENERAL DISORDERS	Chest Pain – Non-Cardiac	3	1.49	0.44-3.52	2	1.09	0.25-2.91
	Chills	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Edema	2	1.09	0.25-2.91	1	0.68	0.1-2.24
	Edema Limbs	2	1.09	0.25-2.91	0	0.28	0.01-1.49

System	AE	CNRT			Varenicline		
	N	N	Estimate	95% CrI	N	Estimate	95% CrI
	Face Edema	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Fatigue	14	5.95	3.45-9.36	17	7.17	4.39-10.83
	Fever	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Flu Like Symptoms	2	1.09	0.25-2.91	5	2.3	0.9-4.68
	General Disorders and Administration Site Conditions - Other, Specify	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Malaise	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Pain	5	2.3	0.9-4.68	2	1.09	0.25-2.91
HEMATOLOGIC	Anemia	0	0.28	0.01-1.49	1	0.68	0.1-2.24
INFECTION	Bladder Infection	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Bronchial Infection	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Kidney Infection	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Otitis Externa	1	0.68	0.1-2.24	1	0.68	0.1-2.24
	Otitis Media	2	1.09	0.25-2.91	1	0.68	0.1-2.24
	Papulopustular Rash	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Pharyngitis	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Rhinitis Infective	0	0.28	0.01-1.49	3	1.49	0.44-3.52
	Sinusitis	2	1.09	0.25-2.91	2	1.09	0.25-2.91
	Upper Respiratory Infection	3	1.49	0.44-3.52	5	2.3	0.9-4.68
	Urinary Tract Infection	0	0.28	0.01-1.49	2	1.09	0.25-2.91
INJURIES	Aortic Injury	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Bruising	2	1.09	0.25-2.91	4	1.9	0.66-4.11
	Fall	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Fracture	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Injury, Poisoning, And Procedural Complications Other	3	1.49	0.44-3.52	3	1.49	0.44-3.52
INVESTIGATIONS	Alanine Aminotransferase Increased	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Weight Gain	3	1.49	0.44-3.52	4	1.9	0.66-4.11
	Weight Loss	0	0.28	0.01-1.49	1	0.68	0.1-2.24
METABOLIC	Appetite Decreased	7	3.11	1.41-5.77	12	5.14	2.84-8.37
	Appetite Increased	17	7.17	4.39-10.83	20	8.39	5.36-12.28
MUSCULOSKELETAL	Arthralgia	1	0.68	0.1-2.24	2	1.09	0.25-2.91
	Arthritis	2	1.09	0.25-2.91	0	0.28	0.01-1.49
	Back Pain	3	1.49	0.44-3.52	5	2.3	0.9-4.68
	Generalized Muscle Weakness	1	0.68	0.1-2.24	2	1.09	0.25-2.91
	Leg Cramps	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Musculoskeletal And Connective Tissue Disorder - Other, Specify	2	1.09	0.25-2.91	0	0.28	0.01-1.49
	Myalgia	5	2.3	0.9-4.68	1	0.68	0.1-2.24

System	AE	CNRT			Varenicline		
	N	N	Estimate	95% CrI	N	Estimate	95% CrI
	Myositis	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Neck Pain	1	0.68	0.1-2.24	1	0.68	0.1-2.24
	Pain In Extremity	2	1.09	0.25-2.91	3	1.49	0.44-3.52
NEOPLASMS	Neoplasms Benign, Malignant and Unspecified	1	0.68	0.1-2.24	2	1.09	0.25-2.91
NEUROLOGIC	Akathisia	1	0.68	0.1-2.24	3	1.49	0.44-3.52
	Dizziness	10	4.33	2.25-7.35	7	3.11	1.41-5.77
	Dysarthria	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Dysgeusia	6	2.71	1.15-5.23	4	1.9	0.66-4.11
	Extrapyramidal Disorder	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Headache	12	5.14	2.84-8.37	20	8.39	5.36-12.28
	Hypersomnia	4	1.9	0.66-4.11	8	3.52	1.69-6.31
	Lethargy	1	0.68	0.1-2.24	6	2.71	1.15-5.23
	Nervous System Disorders - Other, Specify	1	0.68	0.1-2.24	5	2.3	0.9-4.68
	Paresthesia	2	1.09	0.25-2.91	2	1.09	0.25-2.91
	Presyncope	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Somnolence	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Syncope	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Tremor	1	0.68	0.1-2.24	1	0.68	0.1-2.24
PSYCHIATRIC	Agitation	2	1.09	0.25-2.91	2	1.09	0.25-2.91
	Anxiety	10	4.33	2.25-7.35	11	4.74	2.55-7.86
	Concentration Impairment	3	1.49	0.44-3.52	9	3.93	1.97-6.83
	Confusion	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Depression	16	6.77	4.08-10.35	14	5.95	3.45-9.36
	Dreaming Abnormal	35	14.48	10.46-19.23	38	15.7	11.52-20.58
	Insomnia	20	8.39	5.36-12.28	22	9.2	6.02-13.23
	Irritability	5	2.3	0.9-4.68	1	0.68	0.1-2.24
	Psychiatric Disorders - Other, Specify	2	1.09	0.25-2.91	2	1.09	0.25-2.91
	Restlessness	1	0.68	0.1-2.24	0	0.28	0.01-1.49
REPRODUCTIVE	Breast Pain	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Erectile Dysfunction	0	0.28	0.01-1.49	2	1.09	0.25-2.91
	Postmenopausal Bleeding	0	0.28	0.01-1.49	1	0.68	0.1-2.24
	Reproductive System and Breast Disorders - Other, Specify	1	0.68	0.1-2.24	0	0.28	0.01-1.49
RESPIRATORY	COPD	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Coughing	6	2.71	1.15-5.23	3	1.49	0.44-3.52
	Dyspnea	1	0.68	0.1-2.24	0	0.28	0.01-1.49
	Laryngeal Edema	0	0.28	0.01-1.49	1	0.68	0.1-2.24

System	AE	CNRT			Varenicline		
	N	**N**	**Estimate**	**95% CrI**	**N**	**Estimate**	**95% CrI**
	Nasal Congestion	4	1.9	0.66-4.11	0	0.28	0.01-1.49
	Pleural Effusion	1	0.68	0.1-2.24	1	0.68	0.1-2.24
	Productive Cough	2	1.09	0.25-2.91	2	1.09	0.25-2.91
	Respiratory, Thoracic and Mediastinal Disorders - Other, Specify	1	0.68	0.1-2.24	1	0.68	0.1-2.24
	Sore Throat	5	2.3	0.9-4.68	1	0.68	0.1-2.24
	Tickle In Throat	1	0.68	0.1-2.24	0	0.28	0.01-1.49
VASCULAR	Hematoma	2	1.09	0.25-2.91	0	0.28	0.01-1.49
	Hot Flashes	1	0.68	0.1-2.24	3	1.49	0.44-3.52
	Hypertension	3	1.49	0.44-3.52	2	1.09	0.25-2.91
	Hypotension	0	0.28	0.01-1.49	2	1.09	0.25-2.91
	Thromboembolic Event	1	0.68	0.1-2.24	0	0.28	0.01-1.49
IMMUNOLOGIC	Allergic Reaction	0	0.28	0.01-1.49	2	1.09	0.25-2.91

E-Table 6. Number of participants experiencing adverse events in Phase 2 by organ system and treatment. Point Estimates and 95% Credible Intervals are shown. N=number of events; Estimate= estimated rates for N events, based on inverse-probability weighting described in our design.

In Phase 2, we estimated the difference between increased conditions (VAR+/CNRT+) and the respective continue conditions (VAR/CNRT) for both abstainers who stayed on initial treatment and non-abstainers who were re-randomized into it, as well as the two increased conditions vs. each other. There were no adverse events for which these differences exceeded 2% with non-overlapping credible interval. Estimates should be considered descriptive with no correction for multiple comparisons.

System	**Specific AE**	**CNRT**						**Varenicline**					
		CNRT (continue)			**CNRT+**			**VAR (continue)**			**VAR+**		
		n	**Estimate**	**95% CrI**	**n**	**Estimate**	**95% CrI**	**n**	**Estimate**	**95% CrI**	**n**	**Estimate**	**95% CrI**
CARDIOVASCULAR	Chest Pain - Cardiac	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Palpitations	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
DERMATOLOGIC	Pruritus	7	4.7	2.14-8.65	3	6.19	1.88-14.15	3	1.68	0.5-3.97	0	2.08	0.08-10.58
	Rash Acneiform	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Skin Rash	5	3.47	1.36-7.01	1	2.83	0.41-9.09	4	2.14	0.75-4.63	1	5.03	0.74-15.76
EAR, NOSE & THROAT	Allergic Rhinitis	0	0.42	0.02-2.24	0	1.17	0.04-6.06	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Ear Pain	1	1.03	0.15-3.37	1	2.83	0.41-9.09	0	0.32	0.01-1.68	0	2.08	0.08-10.58
GASTROINTESTINAL	Abdominal Pain	0	0.42	0.02-2.24	0	1.17	0.04-6.06	2	1.22	0.28-3.27	2	8.02	1.92-20.23
	Bile Duct Stenosis	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Bloating	1	1.03	0.15-3.37	1	2.83	0.41-9.09	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Constipation	0	0.42	0.02-2.24	1	2.83	0.41-9.09	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Diarrhea	1	1.03	0.15-3.37	0	1.17	0.04-6.06	7	3.51	1.6-6.5	0	2.08	0.08-10.58
	Dry Mouth	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Dyspepsia	2	1.64	0.38-4.36	1	2.83	0.41-9.09	3	1.68	0.5-3.97	2	8.02	1.92-20.23
	Flatulence	1	1.03	0.15-3.37	1	2.83	0.41-9.09	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Gastroesophageal Reflux Disease	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Gastrointestinal Disorders - Other, Specify	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Lower Gastrointestinal Hemorrhage	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Nausea	5	3.47	1.36-7.01	1	2.83	0.41-9.09	11	5.34	2.88-8.85	0	2.08	0.08-10.58
	Oral Dysesthesia	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58

System	Specific AE	CNRT						Varenicline					
		CNRT (continue)			CNRT+			VAR (continue)			VAR+		
		n	Estimate	95% CrI	n	Estimate	95% CrI	n	Estimate	95% CrI	n	Estimate	95% CrI
	Oral Hemorrhage	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Stomach Pain	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Toothache	2	1.64	0.38-4.36	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Vomiting	2	1.64	0.38-4.36	1	2.83	0.41-9.09	0	0.32	0.01-1.68	0	2.08	0.08-10.58
GENERAL DISORDERS	Chest Pain – Non-Cardiac	1	1.03	0.15-3.37	0	1.17	0.04-6.06	3	1.68	0.5-3.97	0	2.08	0.08-10.58
	Edema Limbs	1	1.03	0.15-3.37	1	2.83	0.41-9.09	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Fatigue	3	2.25	0.67-5.28	3	6.19	1.88-14.15	4	2.14	0.75-4.63	2	8.02	1.92-20.23
	Fever	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Flu Like Symptoms	4	2.86	1-6.16	0	1.17	0.04-6.06	1	0.77	0.11-2.53	1	5.03	0.74-15.76
	Injection Site Reaction	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Localized Edema	0	0.42	0.02-2.24	1	2.83	0.41-9.09	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Pain	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
HEMATOLOGIC	Anemia	0	0.42	0.02-2.24	1	2.83	0.41-9.09	0	0.32	0.01-1.68	0	2.08	0.08-10.58
INFECTION	Bladder Infection	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Bronchial Infection	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Herpes Simplex	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Infection	0	0.42	0.02-2.24	0	1.17	0.04-6.06	3	1.68	0.5-3.97	0	2.08	0.08-10.58
	Lung Infection	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Myelitis	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Otitis Media	0	0.42	0.02-2.24	0	1.17	0.04-6.06	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Pharyngitis	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Rhinitis Infective	0	0.42	0.02-2.24	0	1.17	0.04-6.06	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Sinusitis	1	1.03	0.15-3.37	0	1.17	0.04-6.06	3	1.68	0.5-3.97	0	2.08	0.08-10.58
	Skin Infection	0	0.42	0.02-2.24	0	1.17	0.04-6.06	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Tooth Infection	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58

System	Specific AE	CNRT						Varenicline					
		CNRT (continue)			CNRT+			VAR (continue)			VAR+		
		n	Estimate	95% CrI	n	Estimate	95% CrI	n	Estimate	95% CrI	n	Estimate	95% CrI
	Upper Respiratory Infection	1	1.03	0.15-3.37	0	1.17	0.04-6.06	4	2.14	0.75-4.63	0	2.08	0.08-10.58
	Urinary Tract Infection	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Wound Infection	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
INJURIES	Ankle Fracture	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Bruising	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Burn	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Fall	1	1.03	0.15-3.37	0	1.17	0.04-6.06	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Injury, poisoning, and Procedural Complications Other	5	3.47	1.36-7.01	0	1.17	0.04-6.06	2	1.22	0.28-3.27	0	2.08	0.08-10.58
INVESTIGATIONS	Alanine Aminotransferase Increased	2	1.64	0.38-4.36	0	1.17	0.04-6.06	4	2.14	0.75-4.63	2	8.02	1.92-20.23
	Aspartate Aminotransferase Increased	4	2.86	1-6.16	0	1.17	0.04-6.06	6	3.05	1.3-5.89	1	5.03	0.74-15.76
	Blood Bilirubin Increased	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	1	5.03	0.74-15.76
	Creatinine Increased	2	1.64	0.38-4.36	0	1.17	0.04-6.06	4	2.14	0.75-4.63	0	2.08	0.08-10.58
	GGT Increased	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Weight Gain	0	0.42	0.02-2.24	1	2.83	0.41-9.09	3	1.68	0.5-3.97	1	5.03	0.74-15.76
METABOLIC	Appetite Decreased	1	1.03	0.15-3.37	2	4.51	1.06-11.71	6	3.05	1.3-5.89	2	8.02	1.92-20.23
	Appetite Increased	1	1.03	0.15-3.37	1	2.83	0.41-9.09	3	1.68	0.5-3.97	0	2.08	0.08-10.58
	Diabetes Mellitus	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Hypertriglyceridemia	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
MUSCULOSKELETAL	Back Pain	2	1.64	0.38-4.36	1	2.83	0.41-9.09	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Bone Pain	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Generalized Muscle Weakness	0	0.42	0.02-2.24	0	1.17	0.04-6.06	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Leg Cramps	0	0.42	0.02-2.24	1	2.83	0.41-9.09	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Myalgia	2	1.64	0.38-4.36	2	4.51	1.06-11.71	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Myositis	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Pain In Extremity	2	1.64	0.38-4.36	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58

System	Specific AE	CNRT						Varenicline					
		CNRT (continue)			CNRT+			VAR (continue)			VAR+		
		n	Estimate	95% CrI	n	Estimate	95% CrI	n	Estimate	95% CrI	n	Estimate	95% CrI
	Tendonitis	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
NEOPLASMS	Neoplasms Benign, Malignant and Unspecified (Incl Cysts And Polyps) - Other, Specify	3	2.25	0.67-5.28	0	1.17	0.04-6.06	2	1.22	0.28-3.27	0	2.08	0.08-10.58
NEUROLOGIC	Dizziness	2	1.64	0.38-4.36	0	1.17	0.04-6.06	1	0.77	0.11-2.53	2	8.02	1.92-20.23
	Dysesthesia	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Dysgeusia	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Extrapyramidal Disorder	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Headache	7	4.7	2.14-8.65	0	1.17	0.04-6.06	6	3.05	1.3-5.89	1	5.03	0.74-15.76
	Hypersomnia	1	1.03	0.15-3.37	0	1.17	0.04-6.06	3	1.68	0.5-3.97	0	2.08	0.08-10.58
	Migraine	0	0.42	0.02-2.24	0	1.17	0.04-6.06	0	0.32	0.01-1.68	1	5.03	0.74-15.76
	Neuralgia	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Paresthesia	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Presyncope	0	0.42	0.02-2.24	0	1.17	0.04-6.06	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Syncope	0	0.42	0.02-2.24	0	1.17	0.04-6.06	2	1.22	0.28-3.27	1	5.03	0.74-15.76
	Tremor	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
PSYCHIATRIC	Agitation	0	0.42	0.02-2.24	1	2.83	0.41-9.09	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Anxiety	4	2.86	1-6.16	1	2.83	0.41-9.09	4	2.14	0.75-4.63	0	2.08	0.08-10.58
	Concentration Impairment	1	1.03	0.15-3.37	1	2.83	0.41-9.09	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Depression	3	2.25	0.67-5.28	4	7.87	2.81-16.46	8	3.97	1.9-7.1	3	11.01	3.4-24.33
	Dreaming Abnormal	4	2.86	1-6.16	1	2.83	0.41-9.09	3	1.68	0.5-3.97	1	5.03	0.74-15.76
	Dyssomnia	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Insomnia	4	2.86	1-6.16	7	12.93	6.04-22.93	7	3.51	1.6-6.5	0	2.08	0.08-10.58
	Irritability	3	2.25	0.67-5.28	1	2.83	0.41-9.09	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Personality Change	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Psychiatric Disorders - Other, Specify	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58

System	Specific AE	CNRT						Varenicline					
		CNRT (continue)			CNRT+			VAR (continue)			VAR+		
		n	Estimate	95% CrI	n	Estimate	95% CrI	n	Estimate	95% CrI	n	Estimate	95% CrI
REPRODUCTIVE	Irregular Menstruation	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Unintended Pregnancy	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
RESPIRATORY	Bronchospasm	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	COPD	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Coughing	3	2.25	0.67-5.28	1	2.83	0.41-9.09	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Dyspnea	1	1.03	0.15-3.37	1	2.83	0.41-9.09	3	1.68	0.5-3.97	1	5.03	0.74-15.76
	Nasal Congestion	0	0.42	0.02-2.24	0	1.17	0.04-6.06	2	1.22	0.28-3.27	0	2.08	0.08-10.58
	Pleural Effusion	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Postnasal Drip	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Productive Cough	1	1.03	0.15-3.37	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Sore Throat	1	1.03	0.15-3.37	1	2.83	0.41-9.09	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Voice Alteration	1	1.03	0.15-3.37	1	2.83	0.41-9.09	0	0.32	0.01-1.68	0	2.08	0.08-10.58
VASCULAR	Hematoma	1	1.03	0.15-3.37	1	2.83	0.41-9.09	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Hot Flashes	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Hypertension	1	1.03	0.15-3.37	0	1.17	0.04-6.06	3	1.68	0.5-3.97	1	5.03	0.74-15.76
	Superficial Thrombophlebitis	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
IMMUNILOGIC	Allergic Reaction	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
	Autoimmune Disorder	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58
	Lymph Node Pain	1	1.03	0.15-3.37	0	1.17	0.04-6.06	0	0.32	0.01-1.68	0	2.08	0.08-10.58
ENDOCRINE	Endocrine Disorders - Other, Specify	0	0.42	0.02-2.24	0	1.17	0.04-6.06	1	0.77	0.11-2.53	0	2.08	0.08-10.58

E-Table 7. Phase 1 Visit and Medication Compliance

Variable Name	Phase 1 Treatment	
	CNRT (21mg patch+ 2mg lozenge) (N=245)[a]	VAR (Varenicline 2mg) (N=245)[b]
Mean Percent Visit Compliance – Mean (SD)	89(21)	87 (21)
Mean Percent of Medications Taken		
Varenicline - Mean (SD)	85(24)*	87 (22)
NRT-Patch - Mean (SD)	84 (23)	84(23)*
Total NRT-Lozenge – (Count) Median (IQR)	76 (9.75-140)	80(22-135.5)*
a-medication data incomplete or missing for 1/245 **b-medication data incomplete or missing for 3/245** *** Placebo**		

E-Table 8. Phase 2 Visit and Medication Compliance

Phase 1 Treatment	**CNRT**				**VAR 2mg**			
Phase 1 Response	**Abstinent**	**Non-Abstinent**			**Abstinent**	**Non-Abstinent**		
Phase 2 Condition	**CNRT (N=54)a**	**VAR 2mg (N=51)**	**CNRT (N=50)b**	**CNRT+ (N=50)**	**VAR 2mg (N=88)c**	**CNRT (N=41)d**	**VAR 2mg (N=42)e**	**VAR+ (N=39)f**
Variable Name								
Mean Percent Visit Compliance (SD)	90 (19)	86 (25)	78 (32)	78 (31)	92 (22)	72 (36)	77 (31)	79 (33)
Mean Percent of Total Meds Taken								
Varenicline - Mean (SD)	85 (28)*	82 (28)	84 (26)*	84 (26)*	80 (32)	77 (32)*	83 (27)	87 (23)
NRT-Patch - Mean (SD)	80 (28)	77 (31)*	77 (33)	85 (25)	76 (34)*	79 (35)	84 (31)*	85 (30)*
Total NRT-Lozenge - (Count) Median (IQR)	40 (0-118)	65.5 (2.5-135)*	59.5 (0.25-140)	72.2 (0-140)	12 (0-76)*	40 (2.5-130)	68 (3-140)*	40 (0-132)*
Medication data was incomplete or missing for **a=3/54** **b=2/51** **c=7/88** **d=1/41** **e=2/42** **f=1/39** ***Placebo**								

E-Table 9. Phase 2 Outcomes: Absolute Risk Difference (ARD) and 95% Credible Interval for the comparison of each rescue strategy (switch and increase) vs. continuation at each follow-up time point for continuous abstinence among non-abstainers at week 6 within the CNRT and Varenicline (VAR) phase 1 conditions

Phase 1 Treatment	**CNRT**		**VAR**	
Phase 1 Response	**Non-abstainer**		**Non-abstainer**	
Phase 2 Condition	**VAR (switch)**	**CNRT+ (increase)**	**CNRT (switch)**	**VAR+ (increase)**
Follow-up				
EOT+30*				
Continuous Abstinence ARD (95% Crl) Relative to Continue Condition**	6% (3%-10%)	5% (1%-8%)	0% (0%-0%)	8% (5%-11%)
Probability of Non-zero Difference Relative to Continue Condition	>99%	>99%	50%	>99%
6-months* **				
Continuous Abstinence ARD (95% Crl) Relative to Continue Condition	1%(-2%-3%)	3% (0%-6%)	0% (0%-0%)	2% (1%-5%)
Probability of Non-zero Difference Relative to Continue Condition	66%	96%	50%	>99%

* EOT+30 = follow-up point of End of Treatment (EOT at week 12) plus 30 days
** Continuation condition for each respective phase 1 medication (CNRT/VAR) is reference for all comparisons with increase and switch (increase/switch minus continue)
*** 6-months post TQD (target quit-date) follow-up point

E-Table 10. Phase 2 Outcomes: Absolute Risk Difference (ARD) and 95% Credible Interval for the comparison of the switch vs. increase rescue strategies at each follow-up time point for continuous abstinence among non-abstainers at week 6 within the CNRT and VAR phase 1 conditions		
Phase 1 Treatment	**CNRT**	**VAR**
Phase 1 Response	**Non-abstainer**	**Non-abstainer**
Phase 2 Condition	**CNRT+ (increase) vs. VAR (switch)***	**VAR+ (increase) vs. CNRT (switch)**
EOT+30**		
Continuous Abstinence ARD (95% Crl)	-2% (-6%-3%)	8% (5%-11%)
Probability of Non-zero Difference	79%	>99%
6-months***		
Continuous Abstinence ARD (95% Crl)	2% (-1%-6%)	2% (1%-5%)
Probability of Non-zero Difference	89%	>99%

* Switch condition is reference (Increase minus switch)
** EOT+30 = follow-up point of End of Treatment (EOT at week 12) plus 30 days
*** 6-months post TQD (target quit-date) follow-up point

E-Table 11. Phase 2 Outcomes for Phase 1 Abstainers		
	ARD For CNRT vs. VAR	**Probability of Non-zero Difference**
EOT+30 days		
Continuous Abstinence	11% (-1%-22%)	97%
6-months*		
Continuous Abstinence	1% (-11%-12%)	56%
*6-month post TQD=Target Quit-date follow-up point		

E-Table 12 Generalized Estimating Equation (Frequentist) Dynamic Treatment Effect for Each Combined Phase1 and phase 2 pathway.		
Dynamic Treatment Regimen	**Probability of Abstinence at week 12**	**95% Confidence Interval**
*Treatment $_{\text{Phase 1}}$, Treatment $_{\text{Phase 2 if abstinent}}$, Treatment $_{\text{Phase 2 if non-abstinent}}$		
Var, Var,Var	0.21	0.03 0.70
Var,Var,CNRT	0.30	0.04 0.80
Var,Var,Var+	0.42	0.05 0.90
CNRT,CNRT,CNRT	0.19	0.02 0.74
CNRT,CNRT,VAR	0.30	0.03 0.87
CNRT,CNRT,CNRT+	0.31	0.03 0.87

*** Nomenclature:**
Treatment $_{\text{Phase 1}}$ (VAR (Varenicline)/ CNRT) -The treatment applied in phase 1
Treatment $_{\text{Phase 2 if abstinent}}$ (VAR/CNRT)- The treatment applied in Phase 2 for Phase 1 Abstainers
Treatment $_{\text{Phase 2 if non-abstinent}}$ (VAR/CNRT/VAR+/CNRT+)- The treatment applied in Phase 2 for Phase 1 Non-Abstainers

References

1. Cinciripini PM, Tsoh JT, Wetter DW, et al. Combined effects of venlafaxine, nicotine replacement & brief counseling on smoking cessation. *Exp Clin Psychopharmacol*. 2005;13(4):282-292. doi:10.1037/1064-1297.13.4.282.
2. Cinciripini PM, Blalock JA, Minnix JA, et al. Effects of an intensive depression-focused intervention for smoking cessation in pregnancy. *J Consult Clin Psychol*. 2010;78(1):44-54. doi:10.1037/a0018168.
3. Cinciripini PM, Robinson JD, Karam-Hage M, et al. Effects of varenicline and bupropion sustained-release use plus intensive smoking cessation counseling on prolonged abstinence from smoking and on depression, negative affect, and other symptoms of nicotine withdrawal. *JAMA Psychiatry*. 2013;70(5):522-533. doi:10.1001/jamapsychiatry.2013.678.
4. Miller WR, Rollnick S. *Motivational interviewing: Preparing people for change.* 2nd ed. New York: Guilford Press; 2002.
5. Nahum-Shani I, Almirall D, Yap JRT, et al. SMART longitudinal analysis: A tutorial for using repeated outcome measures from SMART studies to compare adaptive interventions. *Psychol Methods*. 2020;25(1):1-29. doi:10.1037/met0000219.
6. Nahum-Shani I, Qian M, Almirall D, et al. Experimental design and primary data analysis methods for comparing adaptive interventions. *Psychol Methods*. 2012;17(4):457-477. doi:10.1037/a0029372.
7. Kidwell KM, Seewald NJ, Tran Q, Kasari C, Almirall D. Design and analysis considerations for comparing dynamic treatment regimens with binary outcomes from sequential multiple assignment randomized trials. *Journal of Applied Statistics*. 2018;45(9):1628-1651. doi:10.1080/02664763.2017.1386773.
8. Ouyang J, Bondell H. Bayesian analysis of longitudinal data via empirical likelihood. *Computational Statistics & Data Analysis*. 2023;187:107785. doi:10.1016/j.csda.2023.107785.
9. Yin G. Bayesian generalized method of moments. *Bayesian Anal.* 2009;4(2). doi:10.1214/09-BA407.
10. Wang, S., Kidwell, K. & Kleinsasser, M. snSMART:Small N Sequential Multiple Assignment Randomized Trial Methods. https://github.com/sidiwang/snSMART. Updated 2023. Accessed January 29, 2024.
11. Wei B, Braun TM, Tamura RN, Kidwell KM. A Bayesian analysis of small n sequential multiple assignment randomized trials (snSMARTs). *Stat.Med*. 2018;37(26):3723-3732. doi:10.1002/sim.7900.
12. Wijeysundera DN, Austin PC, Hux JE, Beattie WS, Laupacis A. Bayesian statistical inference enhances the interpretation of contemporary randomized controlled trials. *J Clin Epidemiol*. 2009;62(1):13-21. doi:10.1016/j.jclinepi.2008.07.006.
13. Goodman SN. Introduction to Bayesian methods I: measuring the strength of the evidence. *Clinical Trials and Meta-Analysis*. 2005;2:282-290.
14. FDA. Innovation/Stagnation: Challenge and Opportunity on the Critical Path to New Medical Products. http://www.fda.gov/oc/initiatives/critical path/whitepaper.html. FDA, 2004. Accessed July 4, 2017.
15. Lilford RJ, Thornton JG, Braunholtz D. Clinical trials and rare diseases: A way out of a conundrum. *BMJ*. 1995;311(7020):1621-1625. doi:10.1136/bmj.311.7020.1621.
16. Spiegelhalter, D.J., Abrams, K.R. & Myles, J.P. *Bayesian Approaches to Clinical Trials and Health-Care Evaluation*. West Sussex, England: John Wiley & Sons Ltd; 2004.